

Dataset Preparation

1. Objective

The AI SOC Copilot requires cybersecurity log data to evaluate its ability to detect threats, extract Indicators of Compromise (IOCs), perform MITRE ATT&CK mapping, calculate threat severity, and generate automated incident response reports. The project supports user-uploaded log files, which are processed dynamically during analysis.

2. Dataset Directory Structure

```
data/
|
|— sample_logs/
|
|— uploads/
|
|— processed/
```

sample_logs

Reserved directory for storing sample security logs used for testing and demonstration purposes.

uploads

Stores log files uploaded by the user through the web interface before analysis.

processed

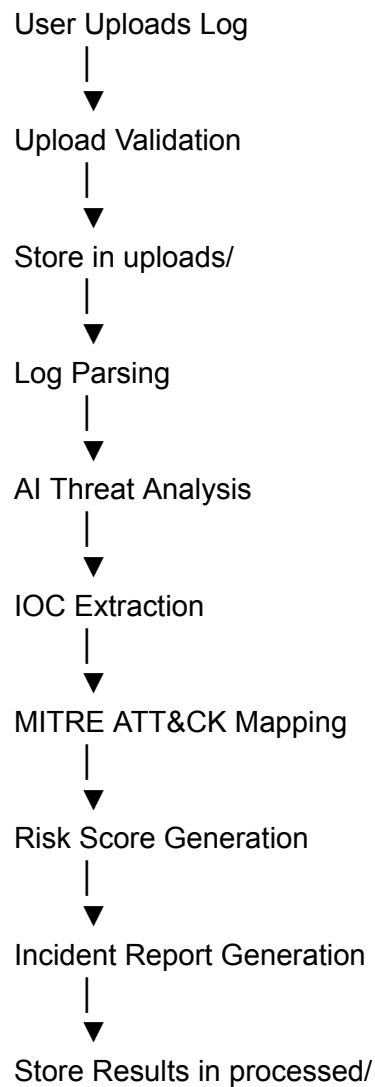
Stores processed outputs generated after AI analysis, including extracted security information used for report generation.

3. Supported Log Formats

The AI SOC Copilot accepts multiple security log formats.

File Format	Purpose
TXT	Windows Event Logs, PowerShell logs, Sysmon logs
JSON	Structured security events
CSV	Bulk security event datasets

4. Data Processing Pipeline



5. Expected Security Events

Security Event	Purpose
User Login	Authentication monitoring
Failed Login Attempts	Brute-force detection
PowerShell Execution	Script abuse detection
Process Creation	Malware analysis
Network Connection	Command-and-Control detection
File Access	Data exfiltration monitoring
Privilege Escalation	Unauthorized access detection

6. Expected Outputs

Input	AI Output
Uploaded Log	Parsed Events
Parsed Events	IOC Extraction
IOCs	MITRE ATT&CK Mapping
MITRE Mapping	Risk Score
Risk Score	Incident Report
Incident Report	PDF Report

7. Conclusion

The prepared dataset workflow enables the AI SOC Copilot to process uploaded security logs, identify malicious activities, extract Indicators of Compromise (IOCs), map adversary behavior to the MITRE ATT&CK framework, assess threat severity, and generate comprehensive incident response reports.